

Madan Bandi

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Summary

Software Engineer with 4+ years of experience in developing scalable applications using Java, Spring Boot, Hibernate, and frontend technologies like React and Angular. Skilled in microservices architecture, RESTful APIs, AWS, Docker, and CI/CD pipelines. Proficient in SQL, MongoDB, and test automation with JUnit. Adept at Agile development and delivering innovative, high-performance solutions.

Technical Skills

Front-end Technologies: HTML5, CSS3, JavaScript, React.js, Angular, jQuery, Redux, AJAX

Database Technologies: MySQL, MongoDB, PostgreSQL, NoSQL, Oracle, Redis, Firebase, Cassandra

Back-end Technologies: Java, Spring Boot, Hibernate, JDBC, J2EE, Node.js, Express.js, Python, Django, ASP.NET, Flask, MVC Framework, SOAP, RESTful APIs, Maven, Micro-services, Bash Script

Tools and Technologies: Git, GitHub, Visual Studio Code, Docker, Jenkins, Kubernetes, AWS (Amazon Web Services), Microsoft Azure, Apache, Nginx, JUnit, Mockito, Swagger, Postman, Apache Kafka, RabbitMQ, GitLab CI/CD, Linux

Work Experience

Software Engineer, PTC

Aug 2023 – Present | Remote, USA

- Developed and deployed a comprehensive e-commerce platform featuring dynamic and intuitive user interfaces, supported by scalable microservices architecture and hosted on AWS Cloud, ensuring high availability and seamless performance.
- Engineered responsive and high-performance user interfaces using React, improving user experience and reducing page load times by 9% through code optimization and state management with Redux.
- Implemented microservices architecture in Java with Spring Boot and Hibernate, enabling seamless integration, data persistence, and independent deployment of RESTful APIs, supported by API documentation using Swagger for scalability and maintainability.
- Optimized AWS infrastructure with Elastic Load Balancing, Auto-Scaling, and Jenkins CI/CD for faster deployments and zero downtime, alongside a robust MySQL database architecture using Hibernate to enhance query performance and data integrity.
- Created a modular application structure with Maven for efficient dependency management and build processes, while using Docker to containerize microservices, standardizing environments and enhancing scalability for seamless deployment.
- Achieved 95% code coverage with JUnit, Mockito, and Postman while collaborating in Agile teams using JIRA and Git, and enhanced performance through regular code reviews and SOLID principles in Java development.

Associate Software Engineer, Legato Health Technologies

June 2021 – Dec 2022 | Bengaluru, India

- Designed and developed scalable RESTful APIs using Java, Spring Boot, and Hibernate, enabling seamless communication between backend services and dynamic front-end interfaces, significantly enhancing user experience and data interaction.
- Optimized structured data management by utilizing advanced SQL and MySQL queries within an AWS RDS database, ensuring high performance, data integrity, and streamlined database operations.
- Ensured software quality and scalability by implementing comprehensive unit testing with JUnit and Mockito, achieving over 95% test coverage and maintaining adherence to industry best practices.
- Automated the software development lifecycle by deploying CI/CD pipelines with Jenkins, with integration of Amazon EC2 for hosting, S3 for artifact storage, and RDS for database management, ensuring rapid, reliable, and error-free software releases.
- Developed authentication and authorization mechanisms using Spring Security and integrated with AWS Cognito, implementing features like user registration, login, and profile management, resulting in improved user engagement and system security.

Software Engineer Intern, Ink wall

Mar 2021 – June 2021 | Hyderabad, India

- Developed the GUI front end for Java applications using Swing, ensuring code quality and best practices while participating in the full software development lifecycle, including requirements gathering, design, coding, testing, and deployment.
- Integrated Java applications with relational databases, writing and optimizing SQL queries, and ensuring data retrieval and manipulation.
- Worked closely with cross-functional teams to gather requirements, design solutions, and implement new features, participating in agile development processes and contributing to continuous improvement initiatives.

Educations

Master of Science, University of South Dakota

May 2024 | Vermillion, SD

Computer Science (GPA: 3.7)

Bachelor of Engineering, Madanapalle Institute of Technology and Science

May 2021 | Angallu, India

Computer Science and Engineering (GPA: 8.35)

Academic Projects

Implementation of Grover's and Shor's Algorithms for Deciphering Encrypted Information

Summer 2023

- Applied Grover's search algorithm for efficient password retrieval from extensive datasets, achieving a quadratic speedup over classical brute force methods.
- Implemented Shor's algorithm to factorize large integers, effectively breaking RSA encryption by deriving private keys through polynomial-time prime factorization.
- Conducted simulations to benchmark the performance of quantum algorithms against traditional cryptanalysis techniques, demonstrating significant computational advantages and scalability improvements.